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Walking—and Running and Jumping—with Dinosaurs and their Cousins, Viewed Through the Lens of Evolutionary Biomechanics

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Synopsis Archosauria diversified throughout the Triassic Period before experiencing two mass extinctions near its end ~201 Mya, leaving only the crocodile-lineage (Crocodylomorpha) and bird-lineage (Dinosauria) as survivors; along with the pterosaurian flying reptiles. About 50 years ago, the “locomotor superiority hypothesis” (LSH) proposed that dinosaurs ultimately dominated by the Early Jurassic Period because their locomotion was superior to other archosaurs. This idea has been debated continuously since, with taxonomic and morphological analyses suggesting dinosaurs were “lucky” rather than surviving due to being biologically superior. However, the LSH has never been tested biomechanically. Here we present integration of experimental data from locomotion in extant archosaurs with inverse and predictive simulations of the same behaviours using musculoskeletal models, showing that we can reliably predict how extant archosaurs walk, run and jump. These simulations have been guiding predictive simulations of extinct archosaurs to estimate how they moved, and we show our progress in that endeavour. The musculoskeletal models used in these simulations can also be used for simpler analyses of form and function such as muscle moment arms, which inform us about more basic biomechanical similarities and differences between archosaurs. Placing all these data into an evolutionary and biomechanical context, we take a fresh look at the LSH as part of a critical review of competing hypotheses for why dinosaurs (and a few other archosaur clades) survived the Late Triassic extinctions. Early dinosaurs had some quantifiable differences in locomotor function and performance vs. some other archosaurs, but other derived dinosaurian features (e.g., metabolic or growth rates, ventilatory abilities) are not necessarily mutually exclusive from the LSH; or maybe even an opportunistic replacement hypothesis; in explaining dinosaurs’ success.

Introduction

Casual observation of a crocodile and a bird might not suggest any profound evolutionary proximity, but Crocodylia and Aves are each other’s closest extant relatives in the vertebrate clade Archosauria. Archosauria originated very late in the Permian Period about 252 Mya and diversified tremendously in the Triassic Period (e.g., Benton 2016). Archosaurs faced the mass extinc-

tions around the Triassic-Jurassic boundary ~201 Mya, leaving only two surviving lineages: Crocodylomorpha, which now includes only Crocodylia as a shadow of its Mesozoic diversity, and Ornithodira (slightly more inclusively, Avemetatarsalia; Nesbitt et al. 2017), including the winged reptiles Pterosauria (extinct by the end of the Cretaceous 66 Mya) along with Dinosauria, which now includes only Aves (Fig. 1). Archosauria

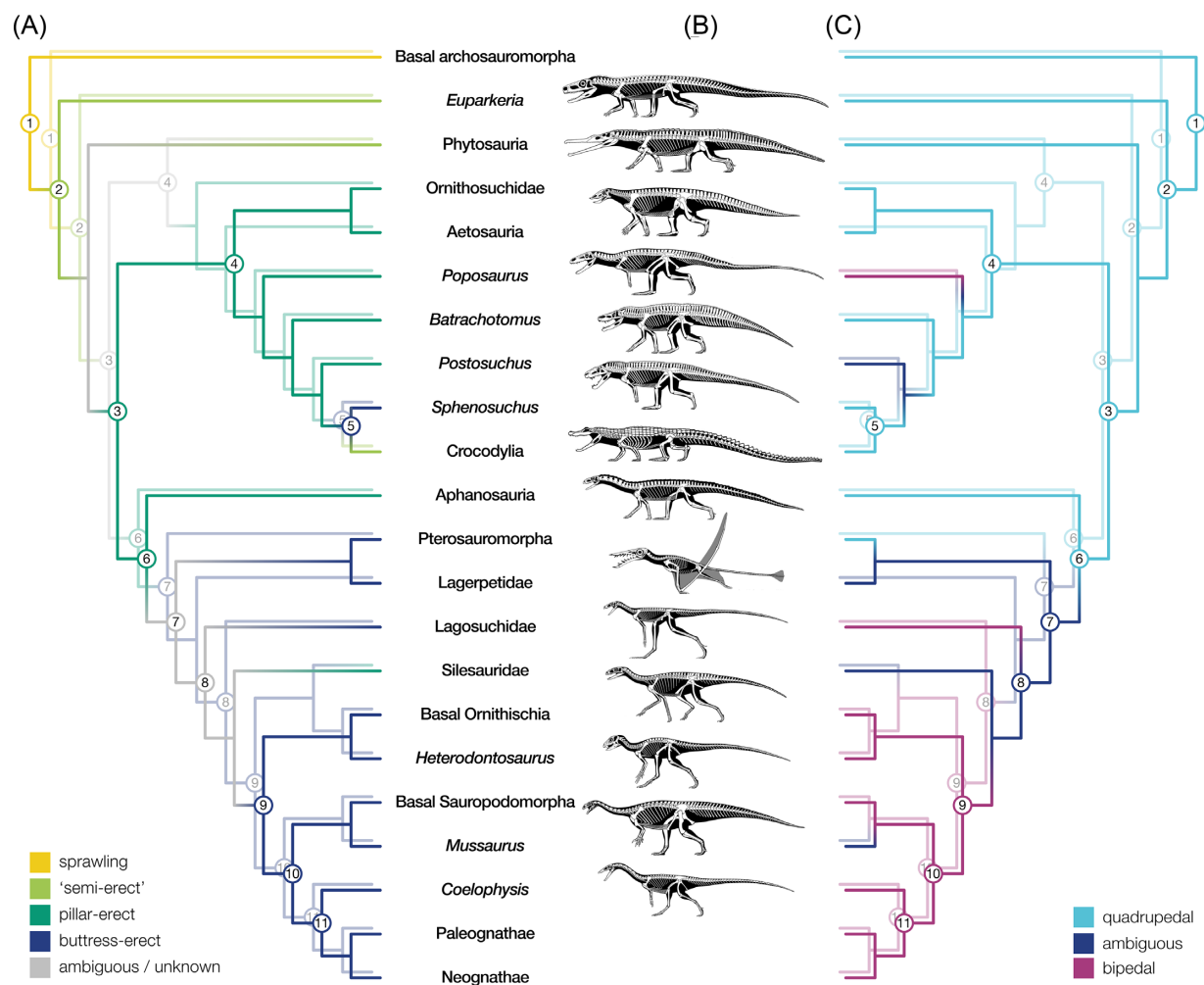


Fig. 1 Postural evolution of Archosauromorpha. **(A)** Evolution of the different hip joint articulations and limb postures superimposed onto a consensus tree (foreground) following Nesbit (2011), Nesbit et al. (2017), von Baczko et al. (2019) and Ezcurra et al. (2020), and alternative tree topology (shadowed) following Ezcurra (2016), Garcia et al. (2019) and Müller and Garcia (2020). “Erect” postures necessitate both relatively parasagittal hip and ankle joints while “sprawling” lack both, and “semi-erect” possess a rotatory ankle joint or lack a relatively parasagittal hip articulation (e.g., Parrish 1986; Demuth et al. 2020; but see Gatesy 1991). **(B)** Skeletal illustrations of various archosauriforms; from top to bottom: *Euparkeria*, *Paleorhinus* (Phytosauria), *Stagonolepis* (Aetosauria), *Poposaurus*, *Batrachotomus*, *Postosuchus*, *Caiman* (Crocodylia), *Teleocrater* (Aphanosauria), *Rhamphorhynchus* (Pterosauriormorpha), *Marasuchus* (Lagosuchidae), *Silesaurus* (Silesauridae), *Heterodontosaurus*, *Plateosaurus* (basal Sauropodomorpha), *Coelophysis*. Drawings not to scale and courtesy of Scott Hartman (<https://www.skeletaldrawing.com/>; except *Euparkeria*, illustrated by Oliver E. Demuth). **(C)** Evolution of bipedalism in archosauromorphs following Grinham et al. (2019), Bishop et al. (2020) and Pintore et al. (2022), superimposed on the same trees as in A. See text for discussion (e.g., *Postosuchus* likely a biped). Clades: 1, Archosauromorpha; 2, Eucrocopoda/Archosauriformes; 3, Archosauria; 4, Pseudosuchia; 5, Crocodylomorpha; 6, Avemetatarsalia; 7, Ornithodira; 8, Dinosauriformes; 9, Dinosauria; 10, Saurischia; 11, Theropoda.

is united as a clade by conspicuous postcranial osteological traits, including a caudoventral portion of the coracoid bearing a tuber, an internal tuberosity of the humerus that is distinctly separated from the proximal articular surface, an olecranon process of the ulna that is prominent but lower than its craniocaudal depth at its base, a lateral “radius” tuber on the proximal ulna, a craniomedial tuber on the proximal femur, a longest metacarpal that is about half the length of the longest metatarsal, and a more caudally oriented calcaneal tuber (~50–90° relative to the transverse plane) (Nesbitt

2011; Ezcurra 2016). These traits are of likely functional relevance to terrestrial locomotion as they suggest enlarged/concentrated attachments for limb muscles or ligaments, greater dominance of the hindlimbs in locomotion (i.e., shortened metacarpals), and leverage for the ankle extensor muscles that is shifted more toward parasagittal function (e.g., Bonaparte 1984; Parrish 1986; Sereno 1991; Sullivan 2010, 2015).

Indeed, archosaurs have long fascinated morphologists and biomechanists because they evolved numerous locomotor specialisations in the Triassic, some of

which extant archosaurs retain. First, the ancestral archosaur had limbs that, to some degree, seem to have been more erect (adducted; drawn closer to the body); unlike the more sprawling posture in earlier tetrapod ancestors. Limb joint morphology along with fossilised trackways together provide strong evidence of this postural shift (Charig 1972; Kubo and Benton 2009), corroborated by the somewhat erect “high walk” that Crocodylia often use, and the quite erect hindlimb posture of Aves. Erect limb posture is strongly associated with a more parasagittal gait (Nyakatura et al. 2019), although it remains uncertain if a highly erect hindlimb posture was ancestral for Archosauria or not (Fig. 1A; Parrish 1986; Sereno 1991; Padian et al. 2010; Sullivan 2015; Demuth et al. 2020). Second, archosaurs repeatedly evolved bipedalism (Fig. 1C); at least once in the crocodile-lineage Pseudosuchia (for example, the bizarre, dinosaur-like Triassic poposaurids; Gauthier et al. 2011; Bates and Schachner 2012; also likely *Smok wawelski* and *Postosuchus* spp.); and at least once (possibly multiple times) in Avemetatarsalia (controversially, pterosaurs; and certainly some or all early dinosaurs and their cousins; Padian 1983, 2012; Novas 1996; Langer et al. 2010, 2013; Pintore et al. 2022). Again, limb morphology and fossilised trackways demonstrate the antiquity of bipedalism in archosaurs (e.g., Brusatte et al. 2011; Kubo and Kubo 2013). Third, foot posture transformed multiple times in archosaurs from the ancestral plantigrade (“flat-footed”) pose to a digitigrade (“tip-toed”) pose (Fig. 1); possibly multiple times in Pseudosuchia (e.g., Crocodylomorpha and poposaurids; Farlow et al. 2014; Schachner et al. 2020; Turner and Gatesy 2021) and once in Ornithodira (Padian 1983; Kubo and Kubo 2016). Digitigrade foot posture has potential benefits for functionally lengthening the limb to increase stride length and perhaps speed (Kubo and Kubo 2012, 2013, 2016), and aligning limb forces with the joints to increase effective mechanical advantage, aiding in support against gravity (Biewener 1989). These and other benefits of digitigrady might also make locomotion more economical, accompanied by less well understood tradeoffs in terms of muscle “gearing” (Carrier et al. 1994). Erect posture and parasagittal gait, bipedalism, and digitigrade foot posture are but three of many remarkable locomotor specialisations of archosaurs—famously, archosaurs also radiated into many habitats by evolving enhanced swimming or climbing abilities or (twice; in pterosaurs and birds) powered flight.

These locomotor specialisations of archosaurs have excited palaeontologists even since the 1800s, leading Richard Owen to coin the name Dinosauria based in part on their apparently advanced locomotion (Owen 1842; Padian 2012). Yet in the 1960s–1970s, new ev-

idence and thinking transformed our view of the importance of archosaur locomotion—and led to debates that continue today. Discoveries of small Triassic forebears of dinosaurs in Argentina helped scientists such as Romer (1971, 1972a, b) and Bonaparte (1969, 1975) to realise that dinosaurs began as small, long-legged (cursorial), perhaps fleet-footed animals; rather than the giants they later became. Bakker and Galton (1974) recognised that early dinosaurs were united by having major muscle attachments on their humerus and femur that should have increased their leverage relative to Triassic Pseudosuchia, and this led them to re-establish the monophyly of Dinosauria, returning to Owen’s (1842) ideas after decades of uncertainty about dinosaur origins. Charig (1972) proposed that there was a sequential “improvement” of limb posture in Triassic archosaurs, from sprawling to “semi-erect” to erect, linked to the ultimate success of dinosaurs and their survival across the Triassic–Jurassic boundary (also see Bonaparte 1984). Parrish (1986) conducted the first real comparative, quantitatively morphometric and “paradigm”-based functional morphology analysis across archosauriforms. He found support for diversity of locomotor function, from more sprawling postures in non-archosaurian archosauriforms and phytosaurs to independent origins of erect gaits in the crocodile lineage, the “crocodile-reversed” ankle-bearing Ornithosuchidae and Dinosauria, inferring that there were competitive advantages of erect postures. Earlier concerns about the possible non-monophyly of Dinosauria were finally fully dismissed as these discoveries and the advent of phylogenetic systematics helped scientists recognise that numerous hindlimb and other synapomorphies unite Dinosauria and its closest relatives (Novas 1996; Langer et al. 2010). These ideas congealed into what can be called the “locomotor superiority hypothesis” (LSH) for dinosaurian evolutionary success. However, that hypothesis soon had one main challenger.

Benton (1983, 1986, 1994) surveyed the global diversity of Triassic–early Jurassic archosaurs and other organisms and inferred that the pattern of originations and extinctions did not match predictions of the LSH (or similar ideas that could be said to centre on differential survival/competition between taxa) that dinosaurs should have increased in diversity while other archosaurs declined during the Triassic. Instead, he favoured an opportunistic (ecological) replacement hypothesis (ORH): as other Triassic species went extinct due to “random” processes (e.g., at the Triassic–Jurassic boundary’s mass extinction), dinosaurs radiated into the vacant ecological space as fortunate “chance” beneficiaries; not superior competitors. Benton’s ideas were expanded by Brusatte et al. (2008a, b), who compiled os-

teological phylogenetic characters into a morphospace analysis of Pseudosuchia vs. Dinosauria/Ornithodira across the Triassic-Jurassic boundary. They proposed that dinosaurs benefitted from “good luck”: starting with low disparity (morphological diversity) and diversity that gradually increased throughout the Triassic/early Jurassic, whereas Pseudosuchia radiated into more disparity and diversity during the Triassic, which crashed by the Jurassic as the clade almost completely “died out by chance.” Hence there was no correlation between Dinosauria/Ornithodira’s rise and Pseudosuchia’s fall from ecological dominance. Further discoveries of possible Early/Middle Triassic dinosaurs (or close relatives) reinforced the notion that dinosaurs were not latecomers to the Triassic (Irmis et al. 2007; Nesbitt et al. 2010, 2013; Brusatte et al. 2011), having coexisted with other archosaurs for many millions of years, in low abundance, until their ascendance late in the Triassic. Thus some consensus was reached that dinosaur origins and gradual diversification early in the Triassic probably had nothing to do with any adaptive superiority they might have had (or not) *versus* other archosaurs (e.g., Brusatte et al. 2010a; Bernardi et al. 2018; Foth et al. 2021).

Soon enough, however, concerns were raised that the ORH or “lucky dinosaurs” might not be so clear-cut of a conclusion. Irmis (2010) countered that many of the potential hypotheses for dinosaur success (*versus* extinction of most Pseudosuchia) were not mutually exclusive, and the hypothesis of chance or opportunistic replacement was not truly a null or default hypothesis, plus that competition or lack thereof was almost impossible to truly test; views broadly echoed by Padian (2012), Langer et al. (2010) and Marsicano et al. (2016). Brusatte et al. (2008a, b, 2010a, b) had likewise acknowledged that dinosaur diversity or disparity patterns were not simple, morphospace trends of phylogenetic characters were not evidence of or against any biotic interactions, and there was some risk of circularity in that these characters were the basis for Pseudosuchia and Ornithodira being recognised as sister clades; thus divergence of disparity between them was almost a foregone conclusion. Nesbitt et al. (2017) used a different phylogeny and more taxa than Brusatte et al. (2008a, b, 2010a, b), obtaining rather different results showing similar morphological disparity in Pseudosuchia (here including Phytosauria; e.g., see Fig. 1) and Avemetatarsalia/Ornithodira throughout the Triassic, demonstrating the importance of these inputs. Foth et al. (2021) found greater disparity in 2D geometric morphometrics of archosaur crania vs. pelves, inferring that the latter might have been more morphofunctionally constrained; yet both anatomical regions exhibited “early burst” high evolutionary rates in the

Triassic. In the meantime, Kubo and Kubo (2012, 2013, 2016) presented a series of studies hinting again at locomotor superiority for Dinosauria: compared with Triassic Pseudosuchia, Triassic dinosaurs overall had longer limbs and stronger digitigrady along with a greater tendency for bipedalism, and fossil trackway evidence suggested this gave them some speed advantages. This research helped clarify that “locomotor superiority” meant only advantages in specific aspects of locomotor performance, such as speed and locomotor cost or efficiency. As lever systems, tetrapod limbs inevitably have biomechanical tradeoffs (e.g., acceleration vs. force production capacity; Biewener 1989; Dick and Clemente 2017) where global superiority of one taxon vs. another is not possible. Benton (e.g., 2016) and colleagues (e.g., Brusatte et al. 2010a; Sookias et al. 2012; Benton et al. 2014; Bernardi et al. 2018) continued to produce studies in favour of the ORH, but did not directly address these critiques and potentially contradictory evidence (yet see Benton 2021).

Here we return to points raised by Irmis (2010) and colleagues, and Kubo and Kubo (2012, 2013, 2016), aiming to reassess the LSH, ORH and other contenders for explaining trends in archosaurian disparity and diversity across the Triassic to the early Jurassic. We do this via a synthesis of understanding, especially studies from the past decade, of the locomotor biomechanics of extant and extinct archosaurs. These studies help address what, if anything, was different between the locomotor functions of Pseudosuchia and Avemetatarsalia/Ornithodira; and how any differences evolved. We begin with a summary of how extant Archosauria (Crocodylia and Aves) move, then examine what the latest studies have taught us about locomotor function, performance and behaviour in the Triassic to early Jurassic archosaurs. Finally, we return to the LSH and ORH with a critical reappraisal and prospectus for these ongoing debates.

Extant Archosauria

The locomotor dynamics of extant archosaurs give hints about what mechanisms might be ancestral for all Archosauria (e.g., Manafzadeh et al. 2021) *versus* which mechanisms divergently evolved in the lineages to Crocodylia or Aves. We begin here with research we have done recently on the terrestrial gaits of Nile crocodiles (*Crocodylus niloticus*), focusing on the “high walk” during steady state locomotion. However, more sprawling limb postures or faster, even asymmetrical, running gaits are common in this and other species of Crocodylia. In those faster gaits, duty factors (= foot-ground contact time divided by total duration of a cycle of footfalls) can get as low as 0.25 and speeds $> 4 \text{ ms}^{-1}$ or

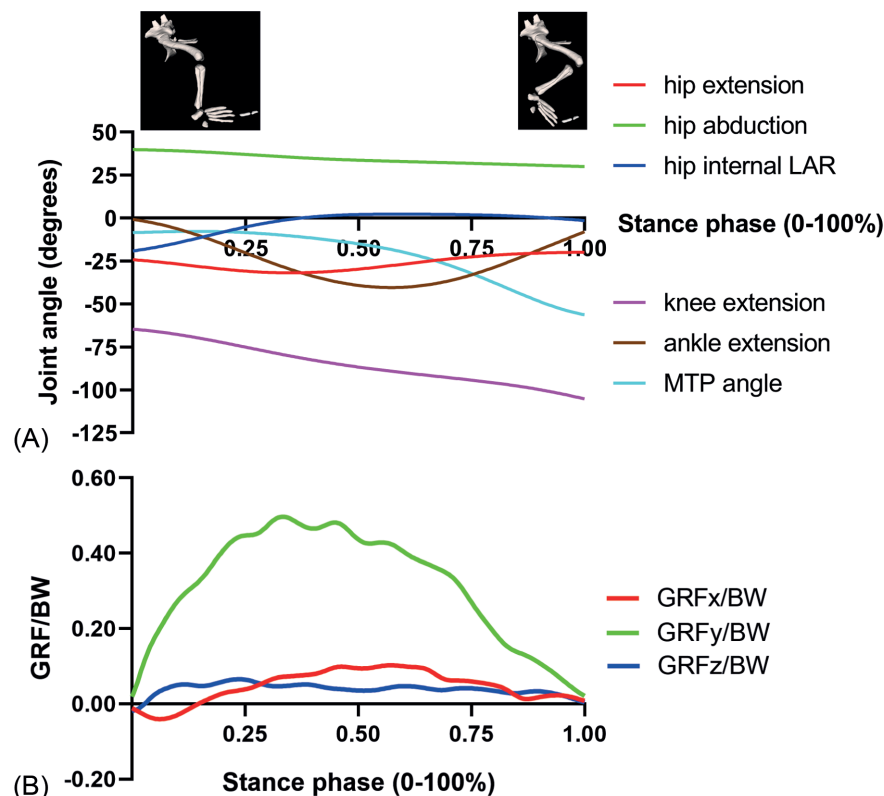


Fig. 2 Three-dimensional hindlimb dynamics of a representative Nile crocodile (*Crocodylus niloticus*) high-walking trial during the stance phase (some data from Wiseman et al. 2021). **(A)** Kinematics: joint angles vs. % stance phase (images of skeleton in lateral view show poses from foot-on to foot-off of right hindlimb); **(B)** Kinetics: ground reaction forces (GRFs) normalised by body weight (BW); GRFx = craniocaudal (horizontal); GRFy = dorsoventral (vertical); GRFz = mediolateral (transverse).

Froude numbers > 10 (Zug 1974; Webb and Gans 1982; Renous et al. 2002; Hutchinson et al. 2019; Wiseman et al. 2021). All experimental protocols were conducted in the Structure and Motion Laboratory of the Royal Veterinary College, via prior approval by the College's Ethics and Welfare Committee (approval number 2016-0089 N) and under a project licence (P0806ABAD) granted by the Home Office (United Kingdom). For ease of comparison with birds, we focus on hindlimb function but some data on forelimb function exist for Crocodylia (e.g., Baier and Gatesy 2013; Otero et al. 2017; Paschenko 2018; Nyakatura et al. 2019). A representative high walk involves a duty factor ~ 0.70 . Each hindlimb primarily generates propulsive (craniad, horizontal) and medial GRFs (Fig. 2A; Table 1). As the crocodile moves forward, the hindlimb retracts, with the hip joint extending (then flexing in later stance), internally rotating around its longitudinal axis and adducting, while the knee has fairly linear flexion and the ankle (dorsi)flexes, then extends (plantarflexes) after mid-stance and the third metatarsophalangeal joint mostly dorsiflexes in late stance (Fig. 2B; Table 1). Intriguingly, our representative trial had similarities to the fourth subject described by Gatesy (1991) that tended

to adduct its hip in stance; unlike the other, more abducted alligators; but our subject did not extend its knee in the same way. These motions and forces otherwise compare favourably with data from high-walking alligators (Gatesy 1991; Blob and Biewener 2001; Reilly et al. 2005; Tsai et al. 2020; Manafzadeh et al. 2021; Iijima et al. 2021), so the general patterns observed in *C. niloticus* appear to be ancestral for Crocodylia. How, then, do these hindlimb dynamics compare with those of birds?

Elegant-crested tinamous (*Eudromia elegans*) are small paleognath birds closely related to ostriches and other traditional ratites. They are ground birds that prefer to run rather than engage in brief bouts of burst flight. Our studies of *Eudromia elegans* (Bishop et al. 2021a, b, c), combined with prior work by Hancock et al. (2007, 2014) and Stoessel and Fischer (2012) established the following locomotor dynamics. In normal walking, these tinamous alternate braking/propulsion, while pushing medially (Fig. 3A; Table 1). Hindlimb retraction during the stance phase consists of the hip extending slightly, possibly internally rotating, and adducting (although the latter varies; e.g., slight abduction in Fig. 3A). Meanwhile, the ankle sometimes remains rather static and the knee and third metatarsophal-

Table 1 Kinematics and kinetics of representative extant archosaur (*C. niloticus* and *E. elegans*) walking trials (see text and Figures 2, 3 for details; data from Wiseman et al. 2021 and Bishop et al. 2021a, c). No statistical comparisons are made. “min” = minimal, “max” = maximal and 0.5 = mid-stance values during stance phase. Kinematics for hip (three rotational degrees of freedom), knee, ankle and “toe”/third metatarsophalangeal (single degrees of freedom) joints. GRF “max1” and “max2” values are for first and second peaks during stance (GRF/BW); and “%stance” is when these occur (0–1 = 0–100%); for all three components of the GRF vector (vertical, craniocaudal, and mediolateral).

Parameter	Crocodylus	Eudromia
Hip extension min	– 32	– 45
Hip extension max	– 20	– 27
Hip extension 0.5	– 30	– 38
Hip abduction min	30	4
Hip abduction max	40	15
Hip abduction 0.5	34	9
Hip internal LAR min	– 19	3
Hip internal LAR max	2	10
Hip internal LAR 0.5	2	9
Knee extension min	– 105	– 102
Knee extension max	– 65	– 42
Knee extension 0.5	– 87	– 75
Ankle extension min	– 40	– 40
Ankle extension max	– 1	– 24
Ankle extension 0.5	– 39	– 31
Toe extension min	– 56	11
Toe extension max	– 8	56
Toe extension 0.5	– 15	27
GRF vertical max1 (BW)	0.50	1.07
GRF vertical max1 (%stance)	0.32	0.16
GRF vertical max2 (BW)	0.48	1.07
GRF vertical max2 (%stance)	0.44	0.68
GRF cranial min (BW)	– 0.04	– 0.14
GRF cranial min (%stance)	0.061	0.10
GRF cranial max (BW)	0.10	0.19
GRF cranial max (%stance)	0.57	0.69
GRF medial min (BW)	– 0.019	– 0.0022
GRF medial min (%stance)	0	0.99
GRF medial max (BW)	0.066	0.15
GRF medial max (%stance)	0.24	0.15

langeal joints show fairly coordinated flexion (dorsi-flexion for the toe), followed by late stance extension (plantarflexion) for the toe (Fig. 2B; Table 1; sometimes accompanied by knee extension, e.g., Kambic et al. 2015). Vaulting (walking) mechanics transitions to bouncing (running) mechanics without an aerial phase at Froude numbers ~ 0.4 – 0.6 , but tinamous can run with aerial phases eventually, reaching speeds $> 2.78 \text{ ms}^{-1}$ and duty factors < 0.39 /Froude numbers > 4 , with

peak vertical GRFs $> 2.5 \text{ BW}$ (Hancock et al. 2007; Bishop et al. 2021c).

The tinamous exhibit some traits that are slightly unusual relative to other small cursorial birds (Movie S1). They hold their toes and tarsometatarsus rotated inwards by 10 – 15° , rather than facing more straight cranially, in standing and while walking and running. The tibiotarsus (knee joint) shows long-axis rotation (LAR) of $\sim 24^\circ$ in external rotation early in stance phase, shifting to about 0° by the end of stance (i.e., internal LAR throughout stance), which decreases with increasing speed. Compared with Helmeted guineafowl (*Numida meleagris*), whose locomotor kinematics are best-studied for birds, tinamous seem to have more hip abduction (the hip adducts thru stance $\sim 9^\circ$ vs. $\sim 3^\circ$ in *Numida*; Kambic et al. 2015), which increases with speed (e.g., 14° in a fast run up to 2 ms^{-1}). We speculate that these kinematic differences in part arise from the unusual foot morphology of *Eudromia*: the first toe (hallux) has been lost, and similar kinematics seem evident in the ostrich *Struthio* (Rubenson et al. 2007), which has also lost its hallux. However, it is possible that all Paleognathae and Neognathae species have derived aspects of their locomotor dynamics in particular ways. Bishop et al. (2021b) used dynamic computer simulations to study vertical jumping in tinamous, estimating that these birds could jump about 0.3 m high (vs. 0.133 standing height), with dynamics and muscle functions that mostly qualitatively matched other birds. The simulated tinamous used a countermovement during pre-launch and landing, respectively with peak vertical GRFs > 2.6 and 4.3 BW ; and jump height performance was most strongly sensitive to ankle extensor muscle force capacity. As it is difficult to get other species such as Crocodylia to jump and impossible for extinct taxa, this predictive simulation approach holds promise for future comparative studies of jumping performance and general dynamics.

Considering the above walking dynamics for Crocodylia and Aves (including other studies by Gatesy 1991; Rubenson et al. 2004, 2007; Willey et al. 2004; Reilly et al. 2005; Kambic et al. 2014, 2015; Schwaner et al. 2022), there are common general patterns for Archosauria studied to date (see also Manafzadeh et al. 2021). Most notably, vertical GRFs have two peaks (much more subtle in Crocodylia) during a stance phase of walking, and the mediolateral GRF vector is medially oriented, of comparable peak magnitude to the horizontal GRF vector (Table 1). Hips extend and internally rotate through stance. The knee shares some similar sagittal motions for archosaurs through stance: extension at least late in stance and possibly flexion earlier in stance; and the toe minimally shares early/mid-stance phase dorsiflexion in archosaurs. The

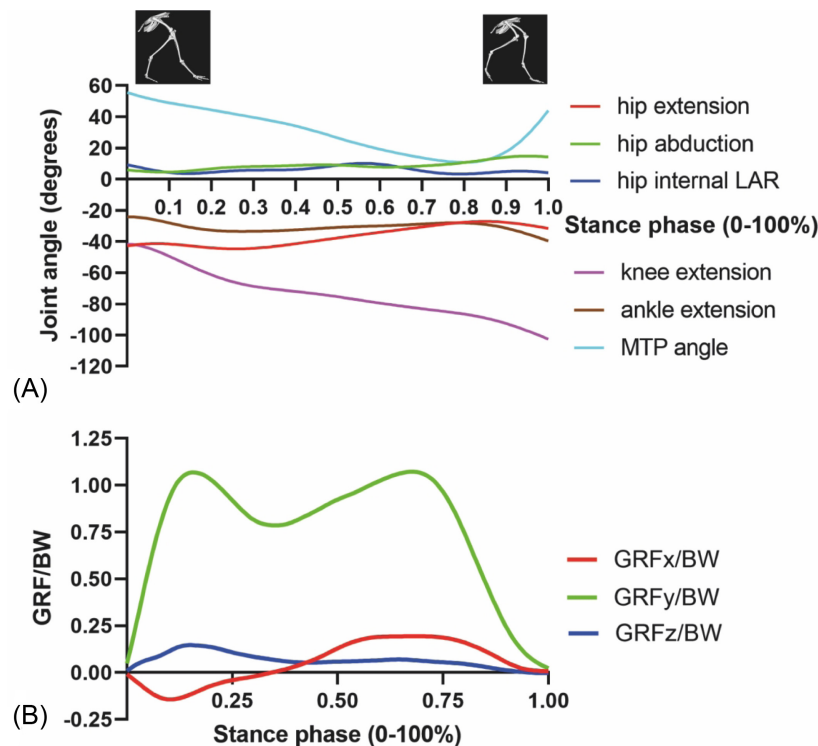


Fig. 3 Three-dimensional hindlimb dynamics of a representative Elegant-crested tinamou (*Eudromia elegans*) walking trial during the stance phase (left hindlimb data from Bishop et al. 2021a, c; reversed to right side to match Fig. 2). **(A)** Kinematics: joint angles vs. % stance phase (images of skeleton in lateral view show poses from foot-on to foot-off); **(B)** Kinetics: ground reaction forces (GRFs) normalised by body weight (BW); GRFx = craniocaudal (horizontal); GRFy = dorsoventral (vertical); GRFz = mediolateral (transverse).

ankle shows extension (sometimes preceded and/or followed by flexion; e.g., Kambic et al. 2015).

However, there certainly are major divergent patterns for extant archosaur locomotor dynamics at similar duty factors for apparently preferred walking speeds. Individual limb GRFs, unsurprisingly, are relatively greater in bipeds (twice as large for vertical GRFs); bipeds must use their hindlimbs to brake and propel rather than specialising them more for propulsion (e.g., Willey et al. 2004); and birds seem to have stronger medially directed GRFs. Birds also show early, asymmetric vertical GRFs rather than having them concentrated around midstance. Variation is present in whether hip abduction (e.g., most crocodylian subjects?) or adduction (e.g., some birds; Rubenson et al. 2007) prevails in stance. Knee motion may be more concentrated on extension in Crocodylia, and the toe tends to have a second flexion cycle late in stance in Aves (vs. Crocodylia), as may the ankle (e.g., Stoessel and Fischer 2012; Kambic et al. 2015), although some birds such as ostriches deviate from those ankle motions (Rubenson et al. 2007). Crocodylia, with their plantigrade pes, tend to emphasize toe dorsiflexion through-

out stance whereas digitigrade birds show some late stance plantarflexion.

So far we have focussed purely on stance (not swing) phase dynamics and joint rotations, not translations (see Manafzadeh et al. 2021, 2022). We will continue to do so in this study, as almost all prior literature on our topic has done so as well. Furthermore, stance phase is arguably simpler to analyse because it imposes severe geometric (the substrate) and biomechanical (supporting the body vs. gravity; as opposed to swinging the limb mainly vs. inertia) constraints. Nonetheless these other phenomena (and others neglected here) certainly deserve further study and integration into our understanding of archosaur locomotor evolution.

It is important to stress, however, that we have overly simplified our treatment of archosaur feet (distal to the ankle joint) above, reducing them to a single (third) toe joint (a weakness of the predictive simulations as noted by Turner and Gatesy (2021) and Turner et al. (2022), Bishop et al. 2021a, b, c). Turner et al. (2020) have revealed detailed 3D kinematics of the feet in extant archosaurs, and a series of studies by Falkingham et

al. (e.g., 2020; also [Falkingham and Gatesy 2014](#)) have integrated kinetic data as well via models and simulations, offering promise of greater insight into 3D foot dynamics even in extinct archosaurs. Together, these studies show that plantigrade archosaurs should tend to exhibit substantial intrapedal 3D motions such as metatarsal spreading, pitching and rolling ([Turner et al. 2020](#)) whereas digitigrade taxa, via their tightly bound, more vertically oriented metapodials, have shared similar phalangeal motions since at least the Early Jurassic Period including “looping” patterns (adduction and flexion coordinated between digits). This reinforces earlier work by [Gatesy et al. \(1999\)](#) showing continuity of theropod dinosaur foot function since the Late Triassic, although the latter study noted that digitigrade early theropods seem to have used foot postures closer to plantigrade (i.e., less vertical than in Aves) early in stance phase. The dynamic simulation research holds much promise for integrating 3D foot form and kinematics with substrate properties to (in)form better animal-ground contact models used in simulations of whole-organism locomotion (extant and extinct). This remains an exciting frontier.

Knowledge of neuromuscular control in relation to locomotor dynamics is vital for better linking locomotor dynamics with morphology (e.g., muscle moment arms; see below) and with methods such as computational models and simulations where control is a necessary input or output parameter. [Gatesy \(1990, 1994, 1999\)](#), [Hutchinson and Gatesy \(2000\)](#) and [Cuff et al. \(2019\)](#) have synthesised data on how neuromuscular control of archosaur hindlimbs during stance phase evolved. While there were some conserved patterns of activation such as “antigravity” extensors (related to conserved joint extension patterns in stance noted above), there are interesting divergences. As [Hutchinson and Gatesy \(2000\)](#) noted, quadrupedal, less erect archosaurs use hip adductor muscles more to support their hindlimbs; whereas bipedal birds use mainly internal rotator muscles. Via simulation studies, [Rankin et al. \(2016\)](#) found that hip abductors were still important in Aves, consistent with Hutchinson and Gatesy’s inference that early bipedal dinosaur(iform)s should have relied on hip abductors for single-leg support, as a transition from adductor to internal rotator-based control mechanisms. [Cuff et al. \(2019\)](#) used experimental electromyographic data and phylogenetic character mapping (following up on [Gatesy 1990, 1994, 1999](#)) to show that archosaurs ancestrally switched their M. iliobtibialis 2 (hip flexor/extensor and abductor, and knee extensor) from swing to stance phase activity. Birds, and possibly earlier dinosaur(iform)s, have switched their M. iliofibularis (hip extensor and abductor, and knee flexor) to stance phase activity as well, and shifted their

M. gastrocnemius pars lateralis (knee flexor and ankle extensor) to start activity just before stance phase, as knee and ankle flexion begin (unlike in typical Crocodylia).

Extinct Archosauria

Recent studies of locomotor function in extinct Triassic Archosauria have focussed either on qualitative characterisation of posture, stance or gait, or (more rarely) quantification of locomotor function. We cover these two types of approaches in sequence here. The first type sometimes has used statistical (e.g., morphometric) methods to categorise “locomotor mode” (e.g., bipedal/quadrupedal stance). Given the multiple origins of some sort of bipedalism in early Archosauria, three recent studies of this type have tried to estimate which taxa were bipedal and how many times or when this capacity (facultative or obligate) evolved. [Grinham et al. \(2019\)](#) used phylogenetic comparative methods and literature-based assessments of bipedal capacity to reconstruct the evolution of bipedalism in Archosauromorpha and found seven origins of obligate bipedalism vs. 14 of facultative bipedalism; only one of the latter being a “transitional” state between quadrupedalism and obligate bipedalism (in the close dinosaur relatives Silesauridae). [Bishop et al. \(2020\)](#) used a combination of morphometric, statistical and biomechanical/functional (based on the body’s centre of mass position) methods to test, based on body/limb form, which archosauromorphs might have been bipedal vs. quadrupedal. They found support for the notion that the pseudosuchian archosaur *Postosuchus*, potentially a close relative of Crocodylomorpha, was bipedal (e.g., [Parrish 1986](#); [Weinbaum 2013](#)). However, they obtained a similar result for the enigmatic *Riojasuchus*, which had been presumed quadrupedal yet some studies have reconstructed it or its close relatives as perhaps facultatively, if not obligately, bipedal ([Walker 1964](#); [von Baczko et al. 2019](#)). Clearly such approaches deserve application to a broader range of archosauromorph taxa and body forms. [Pintore et al. \(2022\)](#) applied 3D geometric morphometrics, including phylogenetic approaches, to the femur of archosauromorphs, testing for shapes typical of bipedal vs. quadrupedal taxa, as well as characterisation of femora into more cursorial/gracile vs. graviportal/robust forms. They obtained categorisations broadly in agreement ([Fig. 4](#)) with those of [Bishop et al. \(2020\)](#), also demonstrating that the femur has a stronger epiphyseal offset, a less curved shaft and a more symmetrical fourth trochanter in quadrupedal archosauriforms than in bipedal avemetatarsalians and pseudosuchians from the Late Triassic. However, they also found some taxa to

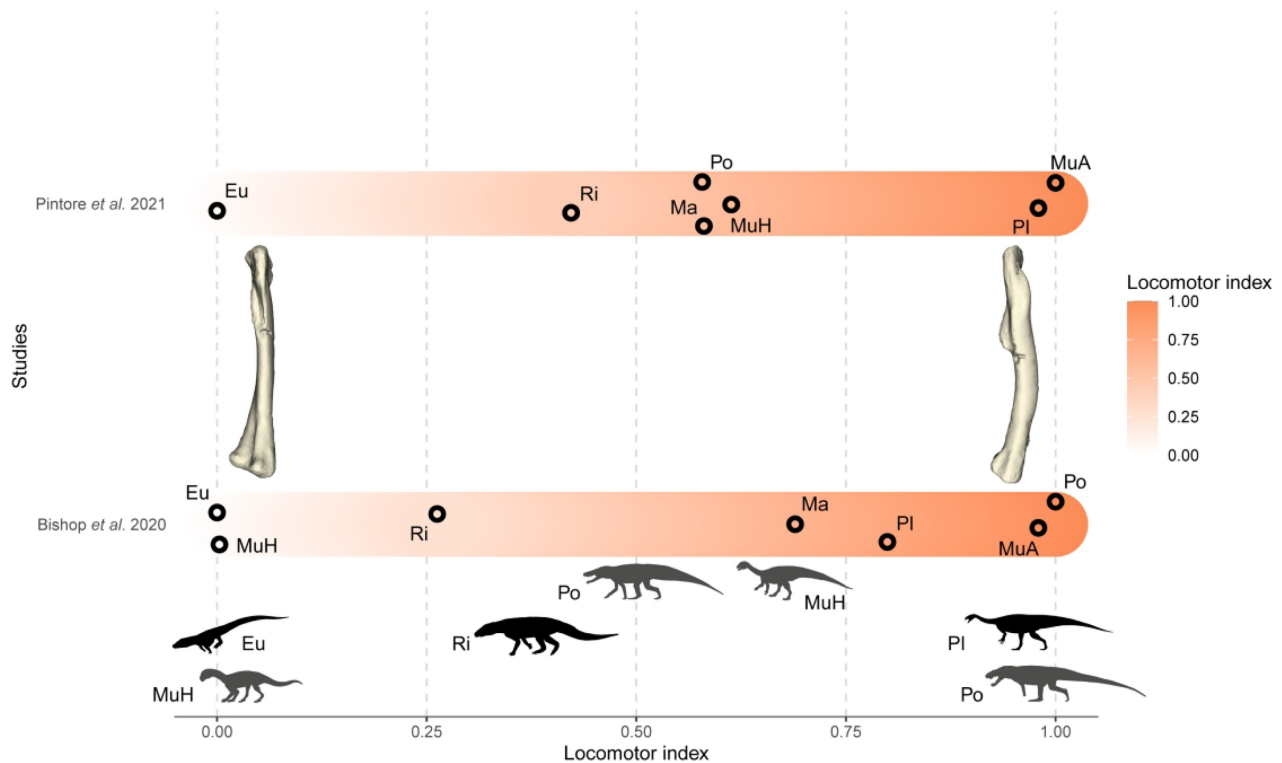


Fig. 4 Comparison of results for the same six archosauriform taxa in [Pintore et al. \(2022\)](#) vs. [Bishop et al. \(2020\)](#). “Locomotor index”: 0 = quadrupedal; 1 = bipedal. Silhouettes in dark grey represent taxa with uncertain locomotor habit. Taxon abbreviations: Eu = *Euparkeria*; Ma = *Marasuchus/Lagosuchus*; Mu = *Mussaurus* (H = hatchling, A = adult), PI = *Plateosaurus*, Po = *Postosuchus*, Ri = *Riojasuchus*. Silhouettes from J. Conway, J. Gonzalez, S. Hartman & A. Otero. Theoretical femoral morphology of quadrupedal (left) and bipedal (right) Triassic archosauriforms from [Pintore et al. \(2022\)](#).

be ambiguous in bipedal/quadrupedal locomotor mode (e.g., some of the dinosaur/pterosaur cousins Lagerpetidae; some Silesauridae; also *Postosuchus* and *Riojasuchus*). As [Pintore et al. \(2022\)](#) determined femur shape to converge on similar forms along with body size in different gracile or robust species, why locomotor mode was not so clearly reflected in femoral shape of some taxa remains to be explained. Nevertheless, disentangling the impact of locomotor mode vs. body size increase in the femoral shape of Triassic archosaurs within a phylogenetic framework broadens our understanding of how locomotor mode might correlate with archosaurian extinctions across the Triassic-Jurassic transition.

Functional morphology is another qualitative (or partly quantitative) approach for categorising locomotor modes or other aspects of function, and has a history almost as lengthy as palaeontology itself. It could be seen as bridging the two types of approaches: the aforementioned statistical/morphometric methods are more pattern-based, seeking common trends in form that correlate with higher-level biological properties, whereas the second, quantitative type is more process-based, seeking or using biological mechanisms and

first principles to infer function using form and other data. Functional morphology certainly uses qualitative form to infer function but may also draw on quantitative morphometric and/or biomechanical data. Most recently in Triassic-Jurassic archosaur functional morphology, [Demuth et al. \(2020\)](#) used 3D digital scans and models of the early archosauriform *Euparkeria capensis* to quantify its hindlimb joint axes and mobility. They found that its hip articulation resembles the “pillar-erect” morphology (e.g., [Fig. 1A](#); [Bonaparte 1984](#); [Parrish 1986](#)), yet the hip joint was fairly mobile except for the most sprawling postures, and its ankle joint had an oblique axis that was inconsistent with a more erect limb posture, so overall they favoured limb poses between very sprawling and erect (mostly consistent with earlier inferences; e.g., [Parrish 1986, 1987](#); [Serenio 1991](#); [Sullivan 2015](#); [Fig. 1A](#)). [Sullivan \(2010\)](#) argued that a lateral tarsal process present in varanid lizards is an analogue of the calcaneal tuber in archosaurs, and used 3D digital models with experimental X-ray measurements of walking *Varanus exanthematicus* to quantify how this process increased ankle extensor leverage (e.g., M. peroneus/fibularis longus). [Granatosky \(2020\)](#) measured the muscle excitation pat-

terns of this and other ankle muscles in the same species via electromyography and confirmed that there was some likely propulsive function. It would be valuable to know how these ankle muscles produce force, torque, work and/or power through a walking stride; so far studies have only tested muscle function more indirectly. Regardless, digital modelling approaches such as Sullivan (2010) and Demuth et al. (2020) have gone beyond simply classifying taxa as quadrupedal/bipedal or sprawling/erect; thereby giving explicit insights into how parts of the skeleton work together, including with muscles.

Bates and Schachner (2012) used very similar 3D modelling methods with the Late Triassic bipedal pseudosuchian *Poposaurus* to quantify its hindlimb muscle moment arms (leverages) and compare them with those of other archosaurs (also see Bates et al. 2015). They found abundant convergence in hip muscle actions with bipedal theropods and other dinosaurs. However, they also noted that *Poposaurus* and *Alligator* (an extant pseudosuchian) were divergent from ornithodiran archosaurs (dinosaurs) in having lower muscle leverage for hip flexion, whereas *Poposaurus* also had low leverage for hip long-axis rotation and abduction/adduction vs. high leverage for ankle extension (from its large calcaneal tuber; ancestral for Pseudosuchia). They made the reasonable inference that these specialisations of hip muscles in pseudosuchians like *Poposaurus* relate to an emphasis of parasagittal limb control linked to that clade's seemingly restrictive "pillar-erect" hip articulations; so modern quantitative approaches reinforced and illuminated more classical inferences from functional morphology. Farlow et al. (2014) focussed on the pes morphology of *Poposaurus*, using a morphometric approach that led them to infer that it was digitigrade; another convergence with ornithodirans. This also concurs with the functional inferences of Turner and Gatesy (2021), who noted that some pseudosuchians with parallel, rather than splayed, metatarsal bones should have been digitigrade; adducing *Poposaurus* as an exemplar.

Allen et al. (2021) used the same kind of 3D musculoskeletal modelling as Bates and Schachner (2012), across a larger sample of archosaurs (but not *Poposaurus*), including the Late Triassic ornithodiran *Marasuchus*. They made the unexpected, tentative discovery that hip medial (internal) long-axis rotation leverage was low (relative to abduction leverage) in ornithodirans (specifically, Dinosauriformes). Like Bates and Schachner's (2012) findings, this result reinforced the idea raised by Hutchinson and Gatesy (2000); using qualitative functional morphology; that early bipedal archosaurs relied on hip abductors for support during stance phase, and only later shifted (within theropod

dinosaurs) to the long-axis-rotation-based mechanism that extant birds use.

Here, we present some previously unpublished data using essentially the same 3D musculoskeletal modelling approach as Bates and Schachner (2012) and Allen et al. (2021), but with a codified method presented by Bishop et al. (2021d) to maximise repeatability and transparency as well as comparability between taxa (also see Gatesy et al. 2022 on joint axes and landmarks for archosaur hindlimbs). We will not detail the methods here, which are fully presented by Bishop et al. (2021d; and see Gatesy et al. (2022)); with extant archosaur examples in Bishop et al. (2021a-c) and Wiseman et al. (2021). Briefly, we: (1) scanned the hindlimb skeletons of eight representative Triassic/Jurassic archosauriforms including *Euparkeria*, *Batrachotomus* and *Poposaurus* (plus 4 extant representatives; Table 2), (2) reconstructed their major hindlimb muscles (Fig. 5A, B) and (3) quantified their leverages, then (4) compared gross patterns across the hip (flexor-extensor) and ankle (extensor-only) joints for major muscle groups. Hip leverages were computed from -65° (flexed) to 45° (extended) hip joint angles (similar to Allen et al. 2021; but -90° to 0° for Aves due to more restricted hip mobility). Ankle leverages were calculated from 0° (dorsiflexed; normally to vertical metatarsus pose but also beyond this to -45° for Crocodylia which use more dorsiflexed angles during stance phase *in vivo*; e.g., Wiseman et al. 2021) to 90° (extended; plantarflexed to plantigrade pose). Leverages were then averaged for simplicity (per taxon, then across clades as needed for comparisons). We chose the hip and ankle because these have the most obvious morphological differences (e.g., pelvis and calcaneal tuber size) that might influence leverage (e.g., Bates and Schachner 2012). Considering the huge body size range of archosauriforms, the issue of dimensional comparison and scaling must be addressed; we divided moment arms by femur length (e.g., Hutchinson et al. 2005; Bates and Schachner 2012) to normalise for body size effects. For simplicity here we assume that using other segment lengths (e.g., effective tibiotarsus, tarsometatarsus) would not alter our results.

We describe qualitative differences in quantitative results of these modelling studies, pending acquisition of a larger sample that would enable more conclusive (e.g., statistical) tests. Hence these findings are preliminary and tentative. Hip extensor moment arms overall did not differ appreciably ($\sim 3\text{--}5\%$ average) for Pseudosuchia (here with *Euparkeria* data included) vs. Dinosauriformes (Fig. 5C: "Pseudosuchia" = *Euparkeria* through *Poposaurus*); unsurprising as the postacetabular ilium can have a similar caudad extent in both

Table 2 Archosauriform musculoskeletal models used here. Body masses are measured for extant archosaurs, or estimated from volumetric models or femoral (and humeral for quadrupeds) circumference following [Campione and Evans \(2020\)](#), but are purely for comparative context here. Femur L = length, used for moment arm normalisation. “Foot pose”: D = digitigrade; P = plantigrade.

Taxon	Specimen #s	Museum	Body mass (kg)	Femur L (m)	Foot Pose	Reference
<i>Euparkeria capensis</i>	SAM PK 6047A + others	Iziko South African Museum	0.8287	0.054	P	Demuth et al. 2020
<i>Batrachotomus kupferzellensis</i>	52970 + others	Staatliches Museum für Naturkunde, Stuttgart	161.2	0.425	P	Bishop et al. 2020
<i>Poposaurus gracilis</i>	YPM 57100	Yale Peabody Museum, New Haven	67.5	0.37	D	(Bates and Schachner 2012)
<i>Crocodylus niloticus</i>	JRHRVC-DDNC06	RVC	2.9	0.067	P	Wiseman et al. 2021
<i>Crocodylus johnstoni</i>	n/a	RVC	20.9	0.112	P	Allen et al. 2021
<i>Marasuchius lilloensis</i>	PVL 3870	Paleontología de Vertebrados, Instituto “Miguel Lillo,” San Miguel de Tucumán	0.288	0.047	D	Allen et al. 2021
<i>Lesothosaurus diagnosticus</i>	NHMMUK RUB 17	Natural History Museum, London	6.3	0.102	D	Allen et al. 2021
<i>Plateosaurus engelhardti</i>	GPIT1 & GPIT2	Eberhardt-Karls-Universität, Tübingen	753	0.57	D	Allen et al. 2021
<i>Mussaurus patagonicus</i>	MLP 68-II-27-I + others	Museo de La Plata, La Plata	1073.3	0.70	D	Otero et al. 2019
<i>Coelophysis bauri</i>	CMNH 10971	Cleveland Museum of Natural History, Ohio	13.9	0.16	D	Bishop et al. 2021cd
<i>Eudromia elegans</i>	JRHRVC-DDT09	RVC	0.534	0.054	D	Bishop et al. 2021a–c
<i>Gallus gallus</i> (junglefowl)	n/a	RVC	2.57	0.105	D	Allen et al. 2021

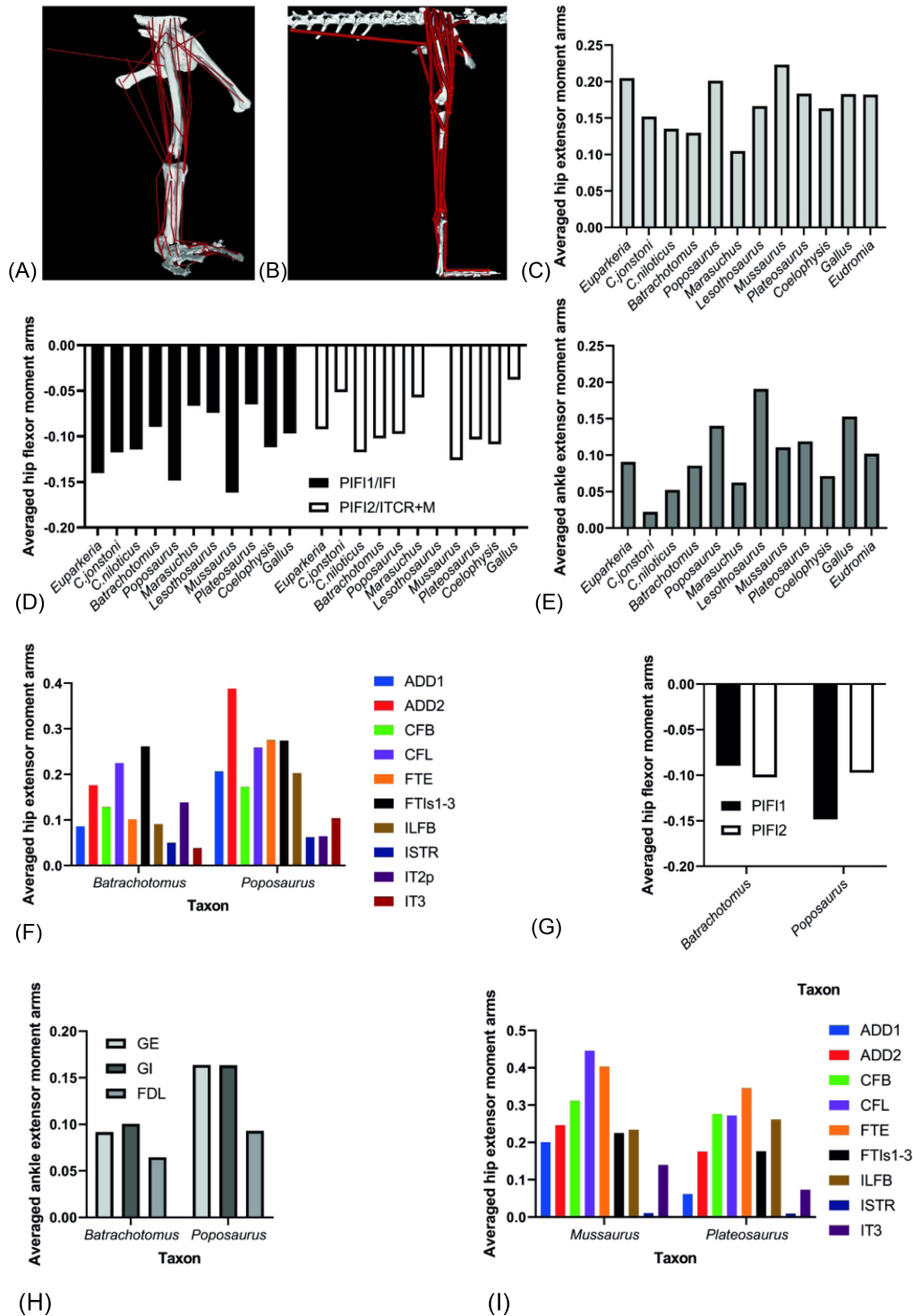


Fig. 5 3D musculoskeletal modelling of archosaur right hindlimbs, with muscle moment arms normalised by femur length. **(A)** *Batrachotomus* and **(B)** *Marasuchus* in lateral view. **(C)** Comparison of averaged hip extensor moment arms (see panels F, I for muscles included). **(D)** Averaged hip flexor moment arms (for PIF1 + 2 muscles—see main text). **(E)** Averaged ankle extensor moment arms: GE = *M. gastrocnemius externus*; GI = *M. gastrocnemius internus*; FDL = *M. flexor digitorum longus*. **(F)** Averaged hip extensor moment arms for individual muscles compared between two Triassic Pseudosuchia. Muscle acronyms: ADD1, ADD2 = *Mm. adductores femoris* 1 + 2; CFB+CFL = *Mm. caudofemorales brevis* + *longus*; FTE + FTIs1-3 = *Mm. flexores tibiales interni externus et interni* 1-3; ILFB = *M. iliofibularis*; ISTR = *M. ischioatrochantericus*; IT2p+3 = *Mm. iliobiales* 2 (posterior part)+3. **(G)** Hip flexors PIF1+2 = *Mm. puboischiofemorales interni* 1 + 2 compared between the same taxa as in F. **(H)** Three ankle extensors (as in E) compared between the same taxa as in F, G. **(I)** Hip extensor moment arms (excluding IT2p; negligible/hip flexor moment arm) compared between two early sauropodomorphs; especially note the large moment arms for *Mm. caudofemorales* (CFB + CFL). For details on muscle homologies see Bishop et al. (2021d).

groups (e.g., [Hutchinson 2001](#)). Yet we found that *Pseudosuchia* had about 30% larger hip flexor moment arms vs. *Dinosauriformes* for the important M. puboischio-femoralis internus 1 + 2 (PIFI1 + 2) deep dorsal thigh muscle group ([Fig. 5D](#)). This difference is evident anatomically from the position of those muscles in *Crocodylia* (well craniad to the hip, especially for PIFI2) and similar inferred positions in most extinct *Pseudosuchia* based on the short preacetabular ilium which has not captured the PIFI1 + 2 origins as occurred in later *Dinosauriformes* (e.g., [Hutchinson 2001](#)). Surprisingly, ankle extensor (plantarflexor) muscles did not emerge as having greater leverage for the pseudosuchians sampled, despite their large calcaneal tubers, which *Dinosauriformes* lack; indeed, the latter lineage tended to have 48% greater ankle extensor moment arms ([Fig. 5E](#)), which might be attributable to their digitigrade foot postures ([Table 2](#)). This preliminary finding deserves deeper sensitivity analyses of the musculoskeletal models in follow-up work with more taxa. These differences are also illuminated by closer comparison of models of two somewhat closely related pseudosuchians: quadrupedal, plantigrade *Batrachotomus* vs. bipedal, digitigrade *Poposaurus* differ by (latter vs. former taxon) 28%, 55% and 90% greater in their hip flexor, hip extensor and ankle extensor leverages respectively ([Fig. 5F-H](#)), likely linked to their different stance and apomorphic pelvis in *Poposaurus* ([Bates and Schachner 2012](#)) despite their similar body sizes. Size-related morphological transformations also alter muscle leverages in complex ways. “Graviportal” archosaurs such as the large early sauropodomorphs *Plateosaurus* and *Mussaurus* have large hip extensor moment arms (~30–210% greater vs. average for *Dinosauriformes*) for M. caudofemoralis longus (and its brevis part) due to its insertion on the distally displaced fourth trochanter (see [Pintore et al. 2022](#)), but other hip extensor leverages are more unexceptional; attributable to their fairly plesiomorphic dinosaurian pelvic morphology ([Fig. 5I](#)). These are just a few examples of how 3D musculoskeletal modelling can reveal basic aspects of archosaur hindlimb biomechanics and evolution.

3D musculoskeletal modelling has also been used, to a lesser degree, to study forelimb function in early archosaurs; and 3D models have been used in more biomechanical, functional studies to infer bipedalism. [Otero et al. \(2017\)](#) modelled forelimb muscles in a crocodile and the early (but already > 1000 kg) sauropodomorph dinosaur *Mussaurus patagonicus*. They inferred that the forelimb joints were unable to put the forelimbs into a fully pronated position for quadrupedalism, and that muscle leverages (and overall functions) in the more sprawling crocodile vs. erect sauropodomorph were heavily influenced by limb pos-

ture. While many forelimb muscles conserved their morphology and presumed functions, others transformed both. For example, M. scapulohumeralis posterior was a supinator of the shoulder joint in *Crocodylus* (and Archosauria ancestrally?) but a pronator in *Mussaurus*; and M. coracobrachialis brevis ventralis was a shoulder extensor vs. flexor, respectively, in those taxa; and M. pronator teres was an elbow extensor vs. flexor. Furthermore, quadrupedalism vs. bipedalism was assessed across ontogeny in *Mussaurus* using 3D digital modelling-based estimates of centre of mass (COM) position ([Otero et al. 2019](#)), following the logic that to stand on one leg a biped must be able to place its COM over its foot. From hatchling to adult *Mussaurus*, the COM shifted caudally (with enlarged tail and reduced head/neck relative size), bringing the COM from a position too far craniad to enable bipedalism to one much more amenable to such behaviour. This study thus supported the inference of [Otero et al. \(2017\)](#) that adult *Mussaurus* were bipedal; but added the insight that this was a secondary ontogenetic transformation. More studies have found similar patterns in other early dinosaurs such as the sauropodomorph *Massospondylus carinatus* ([Reisz et al. 2012](#); but see [Chapelle et al. 2020](#)).

Some of the most recent studies of archosaur locomotion and evolution have used more cutting-edge biomechanical methods that enable deeper, more specific insights into function, performance and behaviour in extinct taxa. [Polet and Hutchinson \(2022\)](#) adopted [Polet's \(2021\)](#) perspective of the “Murphy number” and 2D trajectory optimisation-based optimal control simulations to reconstruct how the quadrupedal early “rauisuchian” *Batrachotomus* may have moved, recognising that it should, like archosaurs ancestrally, have had a large pitch moment of inertia due to its long, heavy body. They tested the validity of their simulation approach vs. empirical data for dogs, finding good agreement. They then compared the predicted limb phase patterns for *Batrachotomus* at different relative speeds to the manus vs. pes footprint positions in fossil trackways assigned to *Brachychirotherium* and *Isochirotherium* (e.g., [Apesteguía et al. 2021](#); [Klein et al. 2006](#); [Petti et al. 2009](#)); both large Middle/Late Triassic ichnotaxa presumed to be “rauisuchians” or roughly similar, heavily built quadrupedal taxa such as aetosaurs ([Lucas and Heckert 2011](#)). [Polet and Hutchinson \(2022\)](#) found that the simulations, in light of the trackways, best supported two gait transitions ([Fig. 6](#)): from a slow walking trot to a “tölt”-like quasi-run with bouncing forelimbs but vaulting hindlimbs, then a full diagonal sequence run with all four limbs bouncing.

Similar optimal control-based simulations have also been used lately to study locomotion of the Late Tri-

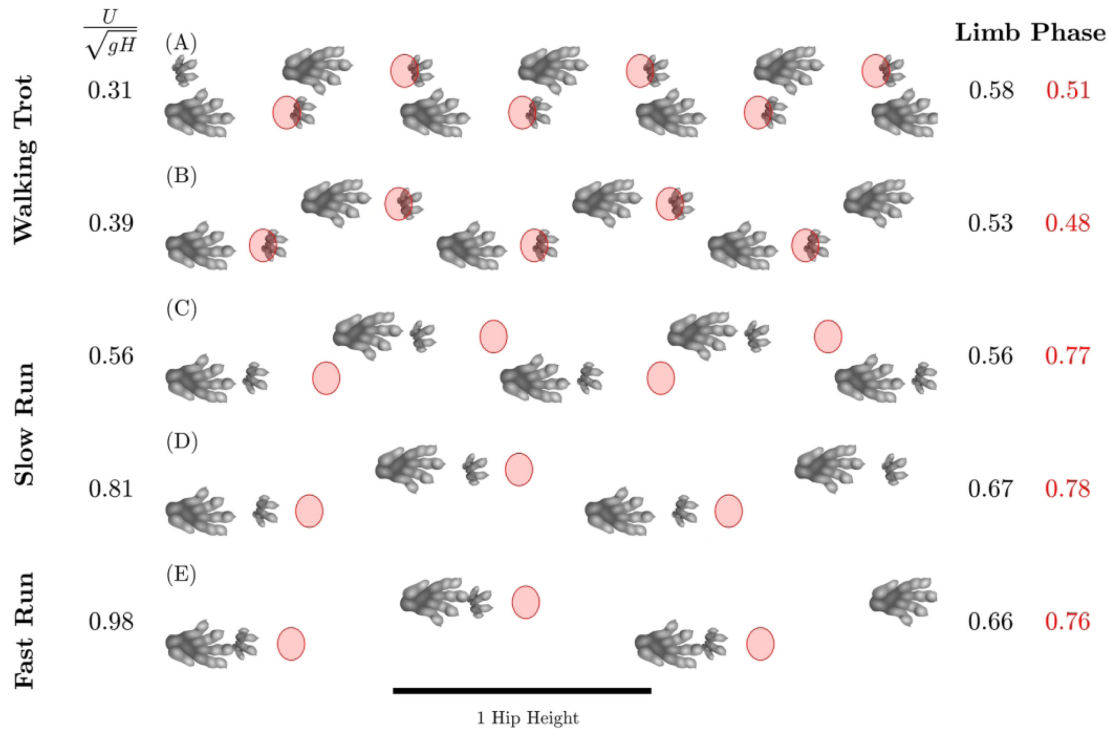


Fig. 6 *Brachychirotherium* trackways with increasing stride length compared to dynamic simulation results from Polet and Hutchinson (2022). At the shortest stride lengths (A, B), the simulation closely matches fossil track phase (distance from pes to manus divided by stride length), and corresponds to a walking trot. To maintain this gait, limb phase must decrease with increased stride length (speed). However, at intermediate stride lengths, track phase becomes constant in the fossil trackways, shown as a static manus position between (C) and (D), and resulting in an increase in limb phase. Simulations predict a shift in gait to a slow run, and an increase in track phase. At the longest stride lengths (D), and therefore fastest speeds, track phase again decreases, representing a maintenance of limb phase, and simulations predict a transition to a fast run. Trackways were reproduced using averaged trackway data from (A, B) Apesteguía et al. (2021) tracks R1 and R2-t5, (C, D) Petti et al. (2009) tracks BsZ-A and BsZ-D, and (E) Klein et al. (2006) trackway NMMNH P-48756. Track width is not representative of width in the fossil tracks. Footprint drawings from <https://commons.wikimedia.org/wiki/File:Brachychirotherium.svg>, used under a CC-BY 3.0 license.

assic theropod dinosaur *Coelophysis* (Bishop 2021a–d). First, Bishop et al. (2021d) used a more static, inverse approach to infer what hindlimb postures it may have used, concluding that more vertical “buttress-like” orientations were favoured. Second, Bishop et al. (2021c) advanced this approach into fully predictive, forward simulations of 3D locomotor dynamics, obtaining similar maximal running speeds to the static approach ($\sim 7 \text{ ms}^{-1}$). They unexpectedly discovered that the tail swung laterally toward each retracting hindlimb during its stance phase, acting to conserve angular momentum like humans swing their arms to do. Additionally, the simulation found muscle excitation and activation patterns supporting earlier conclusions that these early dinosaurs used their hip abductors as important stance phase supportive muscles. Fig. 7 shows the major muscle groups and their activations across a stride. These broadly fall into “stance phase” and “swing phase” active muscles, similar to excitations of homologous archosaurian muscles (Gatesy 1999; Cuff et al. 2019). Yet as Bishop et al. (2021c) noted, quite a few

muscles show secondary bursts of activity in the opposite phase from their main burst (e.g., ITCa, p; IFE; PIFI2; ADD1; FTE; AMB). Similar biphasic or pulsatile bursts also are evident in some extant archosaurian muscles (Gatesy 1997, 1999; Reilly et al. 2005; Cuff et al. 2019).

Muscles in these dynamic simulations tend to be activated in ways that match their apparent optimal actions (i.e., major moment arms; see above). For example, muscles with large hip extensor moment arms such as Mm. flexores tibiales internus et externus, adductores femores and caudofemorales are most active around mid-stance; as are hip abductors/internal rotators such as Mm. iliotrochantericus caudalis, iliofemoralis externus and puboischiofemoralis internus 2. Hip flexors and external rotators/adductors (e.g., Mm. puboischiofemoralis interni 1–3, ischiotrochantericus, puboischiofemoralis internus 1) tend to be more active in swing phase. Muscles with predominantly knee extensor actions (Mm. iliotibialis 2 + 3, femorotibiales) are likewise active around mid-stance whereas knee exten-

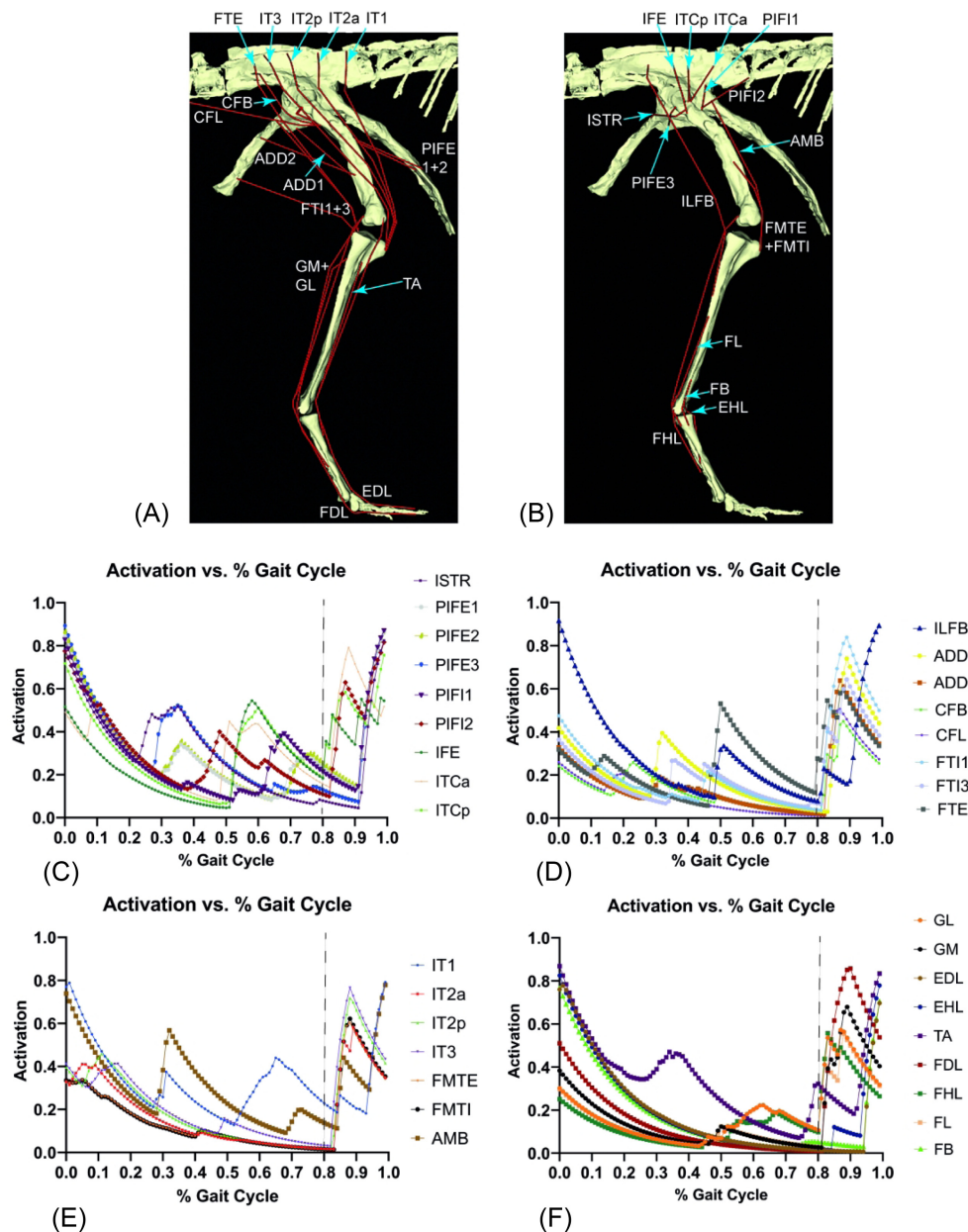


Fig. 7 Biomechanical modelling and simulation of the Triassic theropod dinosaur *Coelophysis* (from Bishop et al. 2021a, c). Superficial (A) and deep (B) right hindlimb muscles in a musculoskeletal model, with muscle acronym labels. (C–F) Muscle activations from the fast-running nominal simulation results from Bishop et al. (2021c); activation levels: 1.0 = 100% active; 0.0 = inactive, plotted against % gait (stride) cycle (0–100% = 0.0–1.0). Muscle acronyms correspond to those in A + B. Vertical dashed lines at 80% stride = start of stance phase (ends at 100% stride). Panels correspond to approximate functional/anatomical groups: C, Hip external rotators/abductors; D, Hip extensors; E, Knee extensors (and hip flexors/extensors); F, Lower limb muscles (mainly ankle flexors/extensors). See Fig. 5 for muscle acronyms plus: AMB = M. ambiens; FMTE+FM TI = Mm. femorotibiales externus + internus; EDL = M. extensor digitorum longus; EHL = M. extensor hallucis longus; FHL = M. flexor hallucis longus; FB+FL = Mm. fibulares brevis + longus; IFE = M. iliofemoralis (externus); ISTR = M. ischiotrochantericus; ITCa, p = M. iliotrochantericus caudalis (anterior, posterior parts); PIFE1–3 = Mm. puboischiofemorales externi 1–3; TA = M. tibialis anterior.

sors (Mm. iliotibialis 1, ambiens) that also have some hip flexor leverage are more active around mid-swing. Lower hindlimb muscles can broadly be differentiated into stance phase ankle extensors/digital flexors (Mm. flexores digitorum longus et hallucis longus, gastrocnemii medialis et lateralis, fibularis longus) vs. swing

phase ankle flexors/digital extensors (Mm. tibialis anterior, extensores digitorum longus et hallucis longus, fibularis brevis). Bishop et al. (2021c) even obtained a late swing phase burst for M. gastrocnemius lateralis that matches electromyographic muscle excitation data for extant archosaurs (Cuff et al. 2019; see above). In-

deed, the general congruence of these results for muscle functions vs. those previously inferred using simpler methods (e.g., general morphology and moment arms; and comparisons with extant taxa) gives some reciprocal confidence to the usage of those methods (e.g., Fig. 5; Bates and Schachner 2012; Allen et al. 2021). Additionally, there are good prospects for harnessing new methods to estimate muscle sizes and incorporate these into dynamic models (e.g., Bishop et al. 2021d; Demuth et al. 2022).

LSH, ORH, or something else?

The above results from recent studies of archosaur locomotion suggest that the LSH for dinosaur success across the Triassic-Jurassic boundary cannot so easily be dismissed, but also that extensive convergent evolution between Pseudosuchia and Ornithodira defies simple scenarios of locomotor superiority. This convergence ranges widely across different levels of biological organisation, including basic skeletal form such as femur shape (Pintore et al. 2022), to muscle functions and leverage (Bates and Schachner 2012; Fig. 5), to limb and foot posture (sprawling-erect continuum; buttress/pillar-erect; plantigrade/digitigrade), and even to behaviours such as quadrupedalism/bipedalism (Otero et al. 2019; Bishop et al. 2020) and perhaps gaits (Polet and Hutchinson 2022). Yet it is timely to also take a fresh look at the fossil record of Archosauromorpha across the Triassic-Jurassic, in terms of other evidence most relevant to the LSH (and ORH). For example, how did body size and metabolism evolve, why did some crocodylomorphs (and pterosaurs) survive along with dinosaurs, how did dinosaur diversity and disparity change, and finally what was the nature of the end-Triassic mass extinctions in terms of overall environmental changes? This final synthesis will enable us to revisit the LSH, ORH, and other hypotheses related to faunal turnover around the Triassic-Jurassic boundary.

Many locomotor parameters depend on (and scale with) body size (i.e., mass; e.g., Biewener 1989; Campione and Evans 2020). In particular, more upright (vertical) limb poses including digitigrade autopodia aid in support against gravity, which can aid in producing the larger forces needed for faster speeds and/or reduce the cost of locomotion by requiring less muscle activation to produce that support (e.g., Dick and Clemente 2017). Hence knowing how body size has evolved is important for reconstructing locomotor evolution. The evolution of early Archosauromorpha seems to show an increase in disparity of body sizes through the Triassic, then a Triassic-Jurassic extinction of larger forms (Turner and Nesbitt 2013; Pradelli et al. 2022; but contradicted by Allen et al. 2019) although perhaps

because of sampling limitations—a common challenge with diversity/disparity analyses; e.g., see Nesbitt et al. 2017). In contrast to this general trend, dinosaurs did not change size disparity around the Triassic-Jurassic boundary (but see Irmis 2010). Turner and Nesbitt (2013) found that dinosaurs increased their range of body sizes while Pseudosuchia decreased theirs; inferring that this pattern was indicative of disruptive selection with an adaptive radiation. Benson et al. (2018) generally agreed, finding evidence for an early burst in ornithodiran body size evolution followed by a Triassic-Jurassic transition with multiple size optima (e.g., larger body size for Sauropodomorpha; vs. Benton et al. 2014). Sookias et al. (2012), however, concluded that increases in dinosaurian body masses during their early evolution were driven by “passive processes,” not Cope’s Rule-like active processes (i.e., selection), but still benefitted from rapid growth rates enabling large maximal body sizes. Intriguingly, small-bodied pterosaurs and early crocodylomorphs survived the mass extinctions, but not their closest relatives, some of which were also small-bodied (e.g., Lagerpetidae). Allen et al. (2019) suggested that there was clade-level selection including against traits such as semi-aquatic habits. Turner and Nesbitt (2013) showed that Crocodylomorpha was the only clade to reduce body size across the Triassic-Jurassic boundary, raising the question of whether pseudosuchians experienced selection against larger body sizes (Pradelli et al. 2022)? The latter study noted that this pattern is combined with a reverse Bergman’s rule in which higher palaeolatitudes evidenced smaller body sizes, perhaps implicating heterothermy and ectothermy rather than homeothermy and endothermy in Triassic Pseudosuchia.

The issue of archosauromorph metabolic strategies is one with a long and complex history of scientific study and we do not attempt to cover it in depth here, much as it is highly relevant to the LSH, ORH and other hypotheses. Studies of ventilatory function and evolution in archosaurs (e.g., Farmer and Sanders 2010; Schachner et al. 2011) are important for placing Dinosauria into a broader context, and perhaps for better understanding why Crocodylomorpha and Pterosauria prospered in the Mesozoic. Dunne et al. (2021) found that Pseudosuchia was more species-rich around warmer equatorial regions, whereas Ornithodira was more cosmopolitan in global distribution; fitting a scenario in which the former were ectothermic and the latter endothermic or mesothermic (also see Mannion et al. 2015; Pradelli et al. 2022). Similarly, Whiteside et al. (2015) inferred that dinosaurs did not become dominant in the tropics of the Triassic Period because their high metabolic rates and endothermy were maladaptive in those unstable “hothouse,” wildfire-prone palaeoenvironments.

Benton (2021) considered archosaurs to have been ancestrally endothermic, key to their survival across the Permo-Triassic boundary (a time of major environmental stresses such as global anoxia). Pontzer et al. (2009) used a biomechanical/metabolic model to assess locomotor physiology. They inferred a link between long (hind)limbs and muscle fascicles, leading to wide aerobic scope (especially in larger-bodied taxa) and greater likelihood of endothermy in dinosauriforms; also consistent with improved endurance and potential for dispersal (see below). Schachner et al. (2011) reconstructed the evolution of the archosaur ventilatory system, inferring a dorsally rigid lung as ancestral for Dinosauria (also see Brocklehurst et al. 2018, 2020). Rapid growth and metabolic rates, advanced ventilatory systems and improved locomotor mechanisms might even be inextricably linked in dinosaurian success: the former two physiological specialisations benefit each other and would be advantageous to fuel widened aerobic scope and locomotor performance (Pontzer et al. 2009).

Arguably, no hypothesis for ornithodiran or dinosaurian success is robust unless the question of why some crocodylomorphs survived the Triassic Period is also resolved. The basic data needed to resolve this are beginning to accumulate at last. As Irmis (2010) noted, Brusatte et al. (2008a, b) excluded Crocodylomorpha from their analyses, thus overestimating the crash of pseudosuchian disparity across the mass extinctions (acknowledged by Brusatte et al. 2010a; Foth et al. 2021). Stubbs et al. (2013) inferred that Triassic crocodylomorphs were “subordinate” to other Pseudosuchia, then became dominant in the Early Jurassic (when there were no large carnivores), “re-visiting” pseudosuchian ecological roles. Furthermore, Crocodylomorpha experienced lower disparity in the Triassic vs. Jurassic (Toljagic and Butler 2013; Foth et al. 2021), partly due to the origin of more aquatic Jurassic lineages. Stockdale and Benton (2021) found correlations between Crocodylomorpha diversity and warm temperatures, concluding that diversity and body size disparity conferred evolutionary opportunities around the Triassic-Jurassic boundary (also hinted at by Dal Corso et al. 2020). Bronzati et al. (2015) instead concluded that the diversification of Crocodyliformes (a less inclusive clade) was linked to the Triassic-Jurassic boundary.

The small, cursorial early crocodylomorphs had their own novelties that might explain their survival into the Jurassic Period. They convergently evolved “buttress-erect” hip joints with dinosaurs (Fig. 1), as well as apparently digitigrade foot postures (e.g., Parrish 1987; Irmis et al. 2013; Turner and Gatesy 2021). Yet it remains far from “fact” (*contra* Benton 2021; also Seymour et al.

2004) that they were bipedal; their locomotor modes are still poorly studied (e.g., Parrish 1986; Kubo and Kubo 2012; Pintore et al. 2022). This leaves claims of Cretaceous “crocodylomorph” bipedal fossil trackways (Kim et al. 2020) highly suspect, as no such trackmakers are known after Late Triassic taxa such as *Poposaurus* and *Postosuchus*. Crocodylomorpha had high evolutionary rates (opposite early dinosaurs) in terms of pelvic 2D geometric morphometrics early in their history (Foth et al. 2021), which surely led to some locomotor and ventilatory novelties. It is generally agreed now that ancestral Archosauria had advanced ventilatory systems including unidirectional airflow and heterogeneous parabronchial lungs with a post-pulmonary septum (Brocklehurst et al. 2020). There is tantalizing evidence that some crocodylomorphs might have had pneumatised postcranial skeletons, but the extent or importance of such pneumaticity is controversial (e.g., O'Connor 2006; Butler et al. 2012; Irmis et al. 2013; Brocklehurst et al. 2020). This is an important issue as it might indicate increased ventilatory capacities, convergent with Ornithodira/Dinosauria, consistent with evidence for somewhat rapid growth (Ricqles et al. 2008; Kubo et al. 2012; Irmis et al. 2013; Legendre et al. 2016; Legendre and Davesne 2020; Klein et al. 2017; Garcia Marsà et al. 2020). If early crocodylomorphs did converge with some ornithodirans in this way, this might even explain their survival into the early Jurassic. It is thought that any such novelties were secondarily lost with aquatic habits in later (Jurassic) Crocodyliformes (Seymour et al. 2004; Ricqles et al., 2008; Legendre et al. 2016; Legendre and Davesne 2020; Benton 2021).

How did dinosaurs evolve? A first, controversial issue is when they evolved; without firmly establishing this it is difficult to resolve how (or why) their diversification relates to that of other lineages. There are undoubted Carnian stage (Late Triassic) trackways (e.g., Carrano and Wilson 2001; Porchetti et al. 2008; Bernardi et al. 2018); firmly consistent with body fossils of true dinosaurs. Purported dinosauriform footprints as old as Early Triassic (e.g., Olsen and Baird 1986; Marsicano et al. 2004, 2007, 2010; Brusatte et al. 2011; Niedzwiedzki et al. 2013; Bernardi et al. 2018; Citton et al. 2020; Marchetti et al., 2021) have been challenged; are they other avemetatarsalians or even non-archosauromorphs (e.g., Padian 2012; Fichter and Kunz 2013; Marsicano et al. 2016; Mancuso et al. 2020)? Diagnostic skeletal remains of Dinosauriformes such as the silesaurid *Asilisaurus* are known from the Middle Triassic (e.g., Sereno and Arcucci 1994; Irmis et al. 2007; Novas et al. 2010; Nesbitt et al. 2010; Langer et al. 2013; Benton et al. 2014; Mancuso et al. 2014; Marsicano et al. 2016). As dinosaurs originated, body

size exhibited complex evolutionary patterns. Body fossils of Ornithischia are rare, and small in body size until the Early Jurassic (Brusatte et al. 2010b; Benson et al. 2018). Theropoda were mostly small-bodied (but increasing in size; Irmis 2010; Turner and Nesbitt 2013), and not very diverse in the Triassic. Sauropodomorpha became abundant and attained very large (>500 kg) body sizes; perhaps benefitting from extinctions of other herbivores in the Late Triassic (e.g., “ecological release” of Langer et al. 2010; from extinction of rhynchosaurs, etc.). Apaldetti et al. (2021) uncovered an expansion of sauropodomorph disparity through the Late Triassic (“early burst” evolution), shifting toward herbivory, seeming unaffected in disparity or diversity by the mass extinctions except that small-bodied taxa went extinct. Hence there are contradictory patterns even just for dinosaur survival into the Jurassic, in which larger-bodied Sauropodomorpha survived vs. smaller-bodied Ornithischia and Theropoda did; and this phenomenon remains unexplained.

The fossil record reveals stepwise extinctions of many archosaurs from the Late Triassic to Early Jurassic (Rigo et al. 2020), but two major events stand out as likely causal mechanisms (e.g., Benton 1994). The Late Triassic’s Carnian stage witnessed the Carnian Pluvial Event (CPE), involving pulses of global increases of humidity and temperature caused by huge volcanic pulses (perhaps four in total, ~234–232 Mya) of the Wrangellia and other large igneous provinces (Dunhill et al., 2018; Lu et al. 2021; Fig. 8). The CPE was roughly concurrent with a phase of dinosaur diversification that seems to have happened during quasi-global aridity in between CPE pulses (Langer et al. 2010; Bernardi et al. 2018; Klausen et al. 2020; Mancuso et al. 2020; Dal Corso et al. 2020; Benton 2021). Hence one major environmental change is correlated with dinosaur success, although this idea is not without challengers (e.g., Dunne et al. 2021). Second, it is now generally accepted that Central Atlantic Magmatic Province (CAMP; Fig. 8) eruptions (~201.5 Mya) caused the final Triassic–Jurassic mass extinction via global warming, elevated greenhouse gases and rising sea levels (He et al. 2020; Petryshyn et al. 2020; Tegner et al. 2020; Wignall and Atkinson 2020; Benton 2021; Fox et al. 2022; Kaiho et al. 2022; Shen et al. 2022), although there may be added complexity (Klausen et al. 2020; Rigo et al. 2020; Ruhl et al. 2011). These changes eradicated coastal lowland ecosystems; a pattern reversed and repeated in a second CAMP phase (Lindström 2021).

Might these major environmental changes give clues to why dinosaurs survived vs. almost all other Archosauriformes did not? Klausen et al. (2020) noted that semi-aquatic Pseudosuchia in coastal areas had “biomechanical inability to cope with flooding” during sea

level changes such as those wrought by the CPE and CAMP; i.e., they could not migrate inland, having limited dispersal ability. In contrast, they pointed out that dinosaurs favoured “marginal ecosystems in the continental interior” (e.g., Brusatte et al. 2011) so dinosaurs did not suffer so much or could recover faster. Furthermore, Klausen et al. (2020) raised the speculation that dinosaur bipedalism may have conferred advantages on land as sea levels changed, although such advantages have not been demonstrated. Yet their general point about Triassic–Jurassic survival fits with a trend noted by Allen et al. (2019): slower dispersers are poor at surviving mass extinctions, and the expected dispersal performance and related locomotor morphology of most Pseudosuchia appears inferior vs. Ornithodira (e.g., Pontzer et al. 2009; Klausen et al. 2020). This generalisation fits with the finding of Dunhill and Wills (2015) that increased geographic range stopped conferring extinction resistance around the Triassic–Jurassic boundary: perhaps, for example, Theropoda survived vs. Phytosauria went extinct because the kinds of environments they favoured and their dispersal abilities, not how broadly distributed they were. If more conclusive methods for testing dispersal ability for different archosaurs were found, that could be more powerful for addressing the LSH vs. other hypotheses.

The literature on archosaurs and the Triassic–Jurassic crisis may see similar data in quite different ways, or at least use different words to describe it (see Irmis 2010). Brusatte et al. (2010b) concluded that dinosaurs likely survived the Triassic “due to differences in growth, metabolism or locomotion” but that these were “not key innovations”; they merely “came in handy”. Benson et al. (2018), however, argued that dinosaurian body size evolution indicated they had “key innovations” behind their early success. Pradelli et al. (2022) likewise wondered if the survival of dinosaurs past the mass extinctions might have been due to selective benefits of their erect bipedal gait, for example. Benton (2014) contended that dinosaur success was not an “accident” but was attributable to a “range of adaptations” advantageous in environmental crises, yet limb posture was “not a factor,” as dinosaurs were bipedal since the Early Triassic. As stated previously, this depends on whether fossil trackways and fragmentary body fossils are accepted as dinosaurian. Langer et al. (2010) surmised that competition around the Triassic–Jurassic boundary might still have played a role in differential survival, and that dinosaurs had “key adaptations” in a changing environment giving them “circumstantial superiority” in the short term during environmental crises; just not throughout the Triassic. Perhaps there is more agreement emerging. Benton (2021) acknowledged Kubo and Kubo’s (2012, 2013) inference that dinosaurs’ long legs

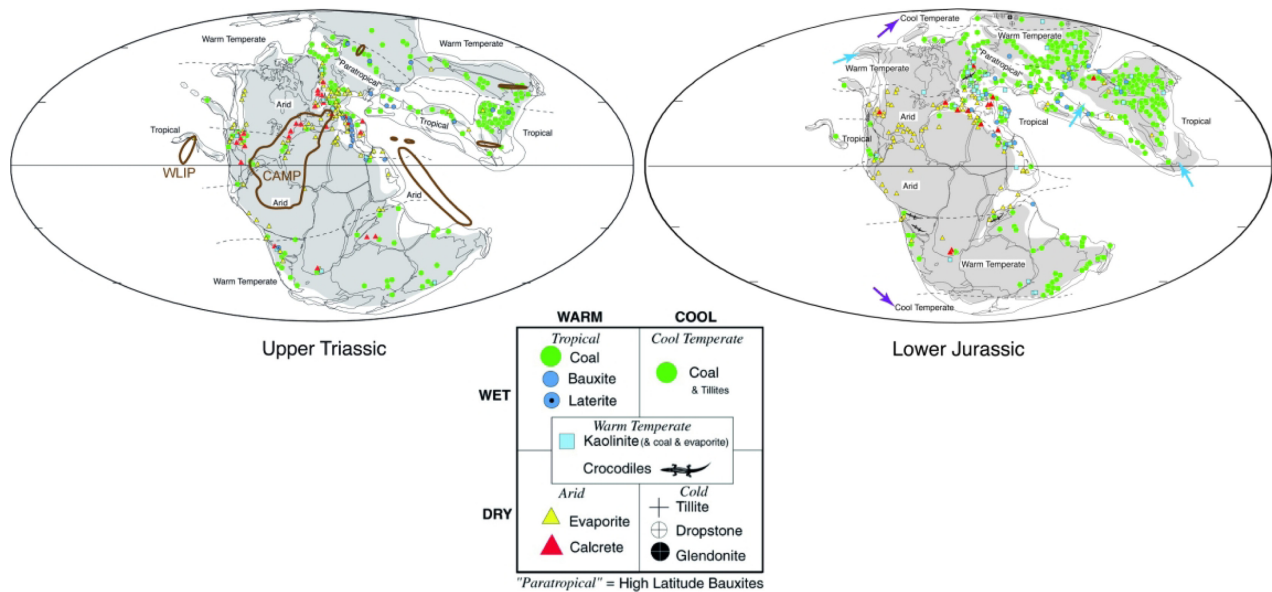


Fig. 8 Upper/Late and Lower/Early Jurassic palaeogeography and palaeoclimate (from Boucot et al. 2013; used with permission). Brown outlines: maximal extents of Wrangel Large Igneous Province (WLIP) and other LIPs (based on Lu et al. 2021) in the Carnian stage, and Central Atlantic Magmatic Province (CAMP; based on Fox et al. 2022) in the Rhaetian stage of Upper Triassic strata. Differences in palaeoclimate are emphasised with purple arrows (“cool temperature” zones absent in Upper Triassic); examples of regions with more exposed land (i.e., lower sea levels) in Lower Jurassic vs. Upper Triassic are emphasised with light blue arrows. Legend shows symbols corresponding to lithological and other evidence for palaeoclimate as per Boucot et al. (2013).

related to long relative stride lengths, lower locomotor costs, and higher stamina; then accepted that, as higher activity levels in the Triassic led to an “arms race,” posture and gait were thus key to dinosaurian success. While dinosaur taxonomic diversity and morphological disparity seem decoupled in the Triassic (e.g., Brusatte et al. 2008a, b, 2010a, b; but see Nesbitt et al. 2017), biomechanical analyses could better test how well morphological vs. functional (e.g., locomotor performance) disparity were coupled (as assumed by Kubo and Kubo 2012, 2013) or not; an issue that is fairly pivotal for conclusively testing the LSH. There seems to be a long way to go yet to fully understand the Triassic–Jurassic transition and archosaurian palaeobiology, but a wide range of perspectives will be valuable in achieving improved understanding. Even if the LSH ends up disfavoured by a strong consensus, it would be exciting to gain confident insight into what the consequences of morphological changes evident in Triassic archosaurs were for higher-level functions and performance, and which of these remain in extant archosaurs as ancient legacies.

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Supplementary data

Supplementary Data available at [ICB](#) online.

Data availability statement

The data underlying this article are available in the article and in its online supplementary material.

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